

MERMAID SVG · PALETTE-BLIND · WCAG AA ASSERTED

Diagram tokens

Per-diagram CSS overrides moved from each palette into `lattice.css`'s **DIAGRAM OVERRIDES** section. Palettes are token-only. Mermaid's `themeCSS` init parameter is gone. Every band/text pair is asserted to clear 4.5:1 in light and dark.

SECTION 01

What changed.

Architecture in one diff.

Before

Each palette file ends with `</em> ===== MERMAID THEME CSS ===== </em>/`, followed by ~540 lines of per-diagram CSS overrides.

`lattice-emulator.js` splits the palette on the sentinel, resolves `var()` to literal hex, strips the `section` prefix, injects via Mermaid's `%%{init: { themeCSS: ... } }%%`.

Tokens carry Mermaid nomenclature: `--mermaid-primary-color`, `--mermaid-pie-purple`, `--mermaid-gantt-active`, ...

After

Each palette declares `--diagram-*` tokens only. No sentinel, no overrides block.

Per-diagram CSS lives in `lattice.css`'s DIAGRAM OVERRIDES section, palette-blind via `var(--diagram-*)`. Loaded by every render path; reaches the inline SVG via host cascade.

No `themeCSS` injection. Mermaid only receives `themeVariables`.

Tokens by role, not by consumer.

Replaces every `--mermaid-<em>` token with a role-named `--diagram-</em>`. A new palette is purely a token-declaration job — no per-palette CSS rules needed.

`--diagram-band-1` .. `--diagram-band-12` — the 12-slot pale-fill cycle for sections, columns, levels, slices, leaves.

`--diagram-band-text-1` .. `--diagram-band-text-12` — paired text, pinned to a fixed dark hex (bands stay pale in both modes; text on top must too). Asserted $\geq 4.5:1$.

`--diagram-stroke` , `--diagram-line` , `--diagram-accent-warm` — structural.

`--diagram-quadrant-N-{fill,text}` — quadrant chart, fill + text paired.

`--diagram-state-{active,done,critical,today,grid}` — gantt task lifecycle.

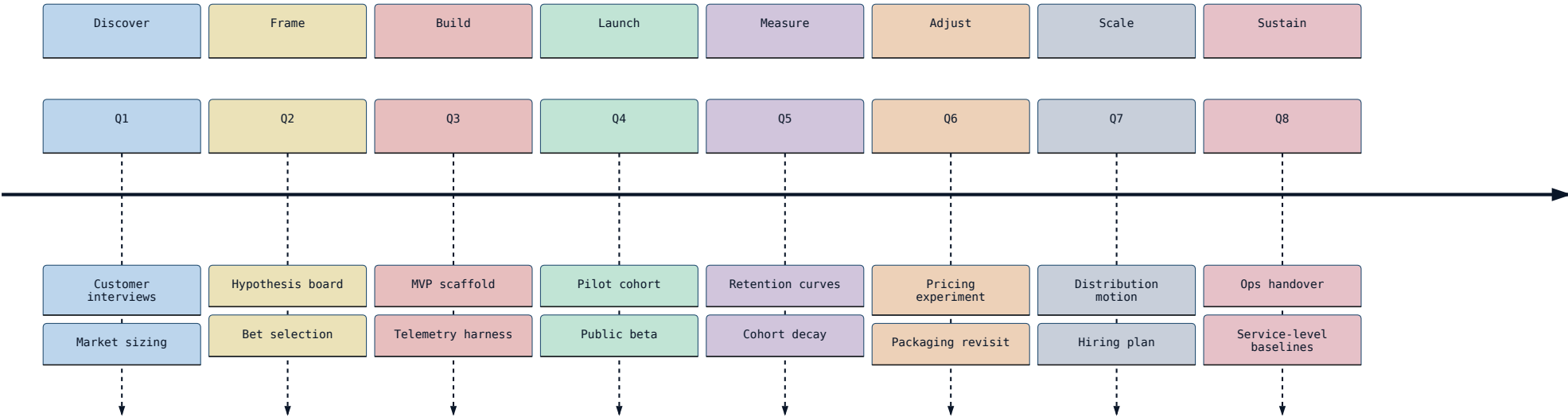
`--diagram-note-{bg,stroke}` , `--diagram-error-{bg,text}` — note and alarm.

`--cat-blue..mauve` stays — categorical mid-tone consumed beyond Mermaid (KPI, chart-family).

SECTION 02

Bands in action.

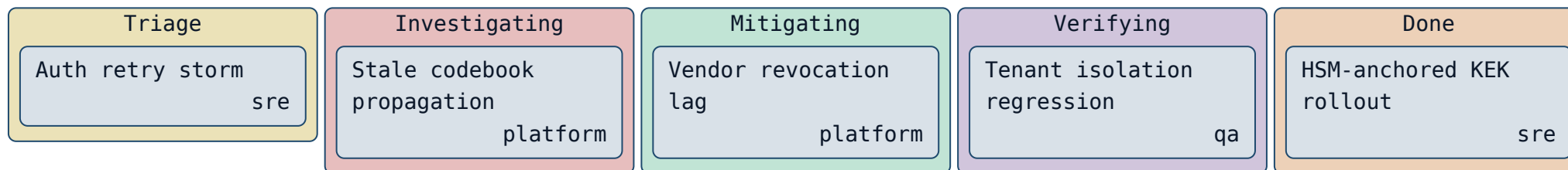
Timeline cycles the band.



KEY INSIGHT

First period reads correctly even though Mermaid emits class `section--1` (the off-by-one quirk in its renderer). Covered explicitly + via `cScaleLabel0..11`.

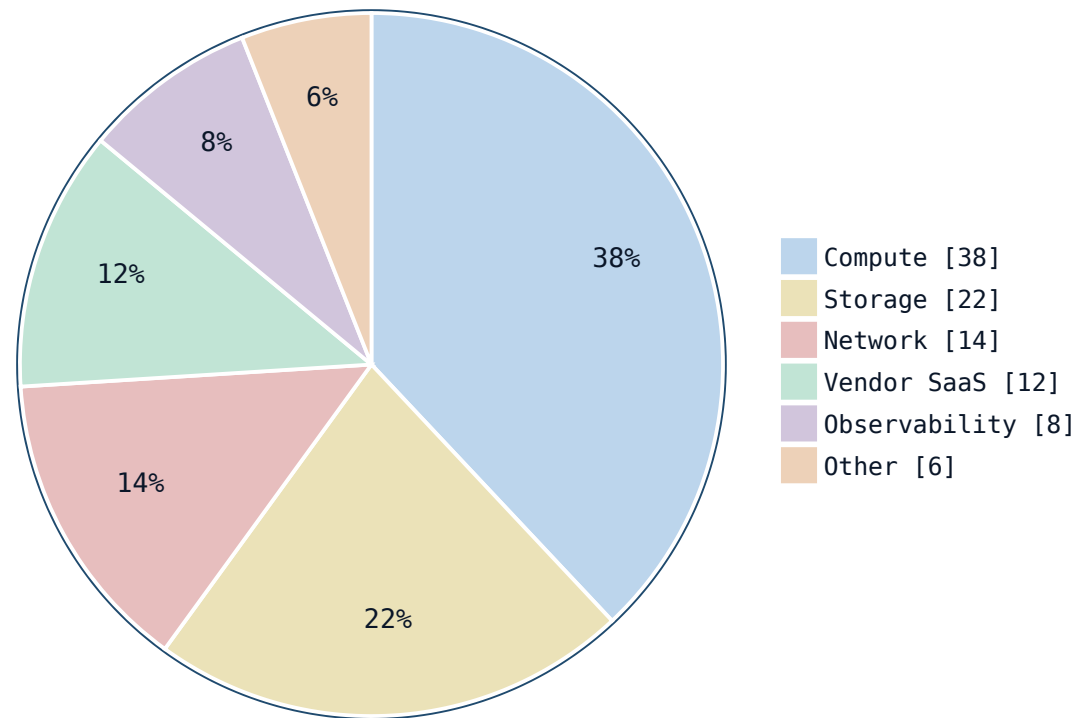
Kanban columns on the band.



KEY INSIGHT

Each column draws from a different `--diagram-band-N`. Tickets sit on `--bg-alt` to read as cards on the swim-lane, distinct from the column hue.

Pie shares the cycle.



KEY INSIGHT

Identical surface contract as kanban; same fills, same paired dark text.

SECTION 03

How it stays honest.

Tested, not asserted.

Two unit-tier checks now ride on every commit. Either failing fails CI.

`test/unit/palette.test.js` — every shipped palette must declare all required `--diagram-<em>` tokens, must NOT carry the legacy MERMAID THEME CSS sentinel, must NOT carry legacy `--mermaid-</em>` token declarations.

`test/unit/contrast.test.js` — for both indaco and cuoio, in both light and dark mode: every band/text pair clears 4.5:1, every quadrant fill/text pair clears 4.5:1, `--text-heading` clears 4.5:1 on `--bg` and `--bg-alt`, `--diagram-error-text` clears 4.5:1 on `--diagram-error-bg`.

If a palette tweak slips below the bar the test fails loudly — lift the text or lift the surface, never lower the bar.

Cleaner palettes, tested contrast.

docs/notes/2026-05-12-diagram-tokens.md

One fewer abstraction. No `themeCSS` parser limits. No Mermaid nomenclature leaking into the palette layer. Every text-on-band pair asserted AA in light AND dark.